

LEDGER · 2026-08-29 · MARINE BIOTECHNOLOGY / COSMETICS

Marine-Safe Biomimetic Sunscreen Filters via Ultrasound-Assisted Extraction of Mycosporine-Like Amino Acids

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Standard Organic UV Filters (Oxybenzone)
Measured advantage	Molar extinction coefficient of 28,100-50,000 M ⁻¹ cm ⁻¹ for isolated MAAs vs ~15,000 M ⁻¹ cm ⁻¹ for Oxybenzone
Time to first revenue	6 months
Gross margin at parity	81.5%
Startup CapEx	\$93,000
Payback from first sale	16.8 months
Capital productivity	1.37x
Regulatory risk	Moderate — qualification costs reported (\$52,000 estimated)

Why it failed

- Voided by: Mismatched comparison

Assessment

The concept fails Sub-test B (Like-for-Like) because the claimed performance advantage relies on the molar extinction coefficient of pure isolated MAAs (Porphyra-334/Shinorine), whereas the commercial product being manufactured is a 17.9% semi-purified crude biomass extract. Additionally, by registering as a cosmetic extract (INCI) to avoid FDA OTC drug approval, it cannot legally perform the primary regulated UV-filtering function of Oxybenzone.

Also flagged

- Price parity asserted, not sourced
- no_production_runsheet
- unsourced_equipment
- no_opex

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-29 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

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